Strategic Capital Allocation in Investment Banking: A Comparative Study of Eight U.S. Firms, 2020 to 2024

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1. ABSTRACT

I studied how eight U.S. investment-banking and advisory firms divided their money between paying shareholders, keeping capital and running the business from 2020 to 2024, and whether these choices were linked to performance in the following year. Panel A has four bank holding companies (Goldman Sachs, Morgan Stanley, JPMorgan Chase, Stifel Financial). Panel B has three advisory boutiques (Evercore, Lazard, Moelis & Company). Jefferies is kept separate as a comparison case. I hand-collected 540 raw values from 40 Form 10-K filings and calculated 70 derived values.

The eight firms returned $233.3 billion to shareholders: $113.9 billion in dividends and $119.4 billion in buybacks. Bank dividends rose over the period, while buybacks changed a lot from year to year. Goldman Sachs, Morgan Stanley and JPMorgan Chase reported standardized CET1 ratios between 13.0% and 17.4%; Stifel's averaged 11.24%. At the boutiques, compensation took 60.7% to 64.0% of net revenue in 2021 and 66.0% to 71.6% in 2023, and operating margins fell from 20.1% to 30.2% in 2021 to between -3.4% and 15.0% in 2023.

In a one-year lagged test, Panel A buybacks had almost no link with next-year return on equity (Pearson r = -0.049, p = 0.858; Spearman rho = -0.035; N = 16). In Panel B, buybacks had a negative but not statistically significant link with next-year operating margin (r = -0.403, p = 0.194; N = 12). My reading is that the same measure can mean different things in a regulated bank and in an advisory boutique. The samples are small, the period is short and unusual, and I have no data on how managers made decisions, so none of these results show cause and effect.

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2. INTRODUCTION

Capital allocation is how a firm decides what to do with the money it earns: put it back into the business, keep it as a buffer against losses, or pay it to shareholders through dividends and buybacks.

I chose investment banking because 2020 to 2024 moved through very different conditions:

1. 2020: the COVID-19 pandemic, followed by government stimulus and emergency Federal Reserve action.
2. 2021: a boom in mergers and acquisitions (M&A), IPOs and debt issuance, with very high fee revenue.
3. 2022 and 2023: the Federal Reserve raised rates by more than 500 basis points, and underwriting and advisory activity fell.
4. 2024: deal activity began to recover, while regulators debated the proposed Basel III Endgame capital rules.

The firms are not built the same way. Bank holding companies take deposits, make loans and must keep regulatory capital under Federal Reserve supervision. Advisory boutiques mainly sell advice and depend on their bankers rather than their balance sheet. So "dollars spent on buybacks" may mean something different in each.

> **Research question**: How do major investment-banking firms allocate capital between business investment, capital retention and shareholder distributions, and how are these patterns associated with later business performance?

This is an observational study. I did not run regressions or try to prove causes; I recorded what the filings show and kept data and opinion apart.

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3. RESEARCH QUESTIONS AND FRAMEWORK

1. **Dividends and buybacks**: How did they change over 2020 to 2024?
2. **Capital and cost flexibility**: What capital ratios did the banks keep, and how did the boutiques' compensation ratios change in the downturn?
3. **Lagged associations**: Were capital-allocation choices in year t associated with performance in year t+1, and did this differ between panels?

I organised my thinking with this chain:

```
Financial conditions (deal cycle, interest rates, regulation, fee revenue)
        |
        v
Decision considerations (NOT observed: risk tolerance, keeping staff, liquidity needs)
        |
        v
Capital-allocation decisions (dividends, buybacks, pay pools, retained capital)
        |
        v
Observable outputs (dividends paid, buybacks, CET1, compensation ratio, margins, ROE)
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This is not a tested model. My data cover only the first and last boxes. A 10-K does not record board discussions or management's reasons, so whenever I talk about decisions it is interpretation, not measured fact.

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4. RESEARCH DESIGN

4.1 Firms and panels

I chose major U.S. investment-banking and advisory firms that reported under U.S. GAAP for all five years. Pooling all eight would mix very different businesses, so I used two panels and one separate comparison firm.

Table 1: Firms and panels

Ticker	Firm	Panel	Business model	Fiscal year-end
GS	The Goldman Sachs Group, Inc.	A	Bank holding company, Category I G-SIB	Dec 31
MS	Morgan Stanley	A	Bank holding company, Category I G-SIB	Dec 31
JPM	JPMorgan Chase & Co.	A	Bank holding company, Category I G-SIB	Dec 31
SF	Stifel Financial Corp.	A	Financial holding company, Category IV	Dec 31
EVR	Evercore Inc.	B	Independent advisory boutique	Dec 31
LAZ	Lazard, Inc.	B	Advisory and asset management (C-Corporation from 2024)	Dec 31
MC	Moelis & Company	B	Independent advisory boutique	Dec 31
JEF	Jefferies Financial Group Inc.	Separate	Broker-dealer and investment bank	Nov 30

Jefferies is outside both panels because its year ends on 30 November and it falls under SEC broker-dealer capital rules (Rule 15c3-1), not Federal Reserve bank holding company rules. It appears in the revenue and payout tables but not in the lagged tests.

4.2 Period and methods

The study covers fiscal years 2020 to 2024, giving four transitions: 2020 to 2021 (pandemic to boom), 2021 to 2022 (peak to rate rises), 2022 to 2023 (weak underwriting and banking stress) and 2023 to 2024 (early recovery).

1. **Descriptive analysis**: levels and changes in revenue, payouts, capital and margins for each firm.
2. **Lagged analysis**: I paired a capital-allocation measure in year t with a performance measure in year t+1 and calculated the Pearson correlation (r) and the Spearman rank correlation (rho). Spearman uses ranks, so it is less affected by one or two extreme values, which matters in small samples.

4.3 Variables

I collected revenue, payout (dividends, buybacks), capital (CET1, Tier 1 leverage and supplementary leverage ratios), performance (ROE, ROTCE, efficiency ratio, operating income and margin, compensation ratio) and balance-sheet (cash, funded debt) variables. Appendix A gives the definitions.

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5. DATA AND PROVENANCE

All data come from the firms' Form 10-K reports on the SEC's EDGAR system; I did not use commercial databases. The dataset covers 40 filings (8 firms x 5 years) with 540 raw values in `PHASE3_RAW_DATA.csv`, taken from the income statement, balance sheet, cash-flow statement and the capital resources section of Item 7 (MD&A).

I calculated 70 derived values (`PHASE3_DERIVED_DATA.csv`):

- Total cash returned = cash dividends paid + share repurchases
- Shareholder distribution ratio = total cash returned / net revenue
- Operating margin = operating income / net revenue (Panel B)
- Net liquid cushion = cash and cash equivalents - funded long-term debt (Panel B)

Share repurchases are the cash paid for buying back common stock in the cash-flow statement. Following my data dictionary, this includes shares taken back to cover employees' tax when stock awards vest, so it is not only open-market buybacks.

Each raw value has a status label: **DIRECT** (450 values, reported in the 10-K); **NOT_APPLICABLE** (20, the measure does not exist for that firm, for example Stifel's SLR and Jefferies's CET1; not treated as zero); **STRUCTURALLY_NON_COMPARABLE** (50, the cash and debt of the banks and Jefferies, which are part of their lending and trading business and so not compared with the boutiques'); **DERIVED** (15, the boutiques' operating margins); and **UNAVAILABLE** (5, Moelis's technology spending). Each value is linked to its filing in the provenance files (Appendix D). While revising this paper I rechecked the numbers in the text against the data files and corrected several errors in the earlier version.

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6. DESCRIPTIVE FINDINGS

6.1 Revenue

All eight firms had higher net revenue in 2024 than in 2020, from +18.6% at Lazard to +55.9% at Stifel (Table 2, Figure 3).

Table 2: Net revenue, 2020 to 2024 (USD millions)

Ticker	Panel	2020	2021	2022	2023	2024	Change 2020 to 2024
GS	A	44,560.0	59,339.0	47,365.0	46,253.0	53,510.0	+20.1%
MS	A	48,197.0	59,755.0	53,668.0	54,187.0	61,814.0	+28.3%
JPM	A	119,543.0	121,649.0	128,695.0	158,104.0	170,154.0	+42.3%
SF	A	3,817.8	4,783.1	4,592.8	5,159.3	5,951.7	+55.9%
EVR	B	2,285.3	3,307.1	2,778.9	2,442.7	2,996.4	+31.1%
LAZ	B	2,646.8	3,273.8	2,855.1	2,593.2	3,139.9	+18.6%
MC	B	943.3	1,540.6	985.3	854.7	1,194.5	+26.6%
JEF	Separate	6,880.4	8,945.5	7,149.3	7,441.4	10,515.1	+52.8%

JPMorgan Chase grew from $119.5 billion to $170.2 billion (+42.3%). I think part of this came from higher net interest income after the Federal Reserve's 525-basis-point rate rises and part from its purchase of First Republic Bank in May 2023, but my data do not separate the two.

Investment-banking revenue peaked in 2021 at every firm (Figure 1), for example $14,877M at Goldman Sachs, $13,216M at JPMorgan Chase, $3,240M at Evercore and $1,540.6M at Moelis. By 2023 it had fallen to $6,218M, $6,519M, $2,375M and $854.7M respectively. In 2024 the boutiques' advisory revenue was above its 2020 level (+31.2% at Evercore, +25.7% at Lazard, +26.6% at Moelis), while investment-banking revenue at Goldman Sachs (-17.9%), Morgan Stanley (-12.6%) and JPMorgan Chase (-6.1%) was still below 2020.

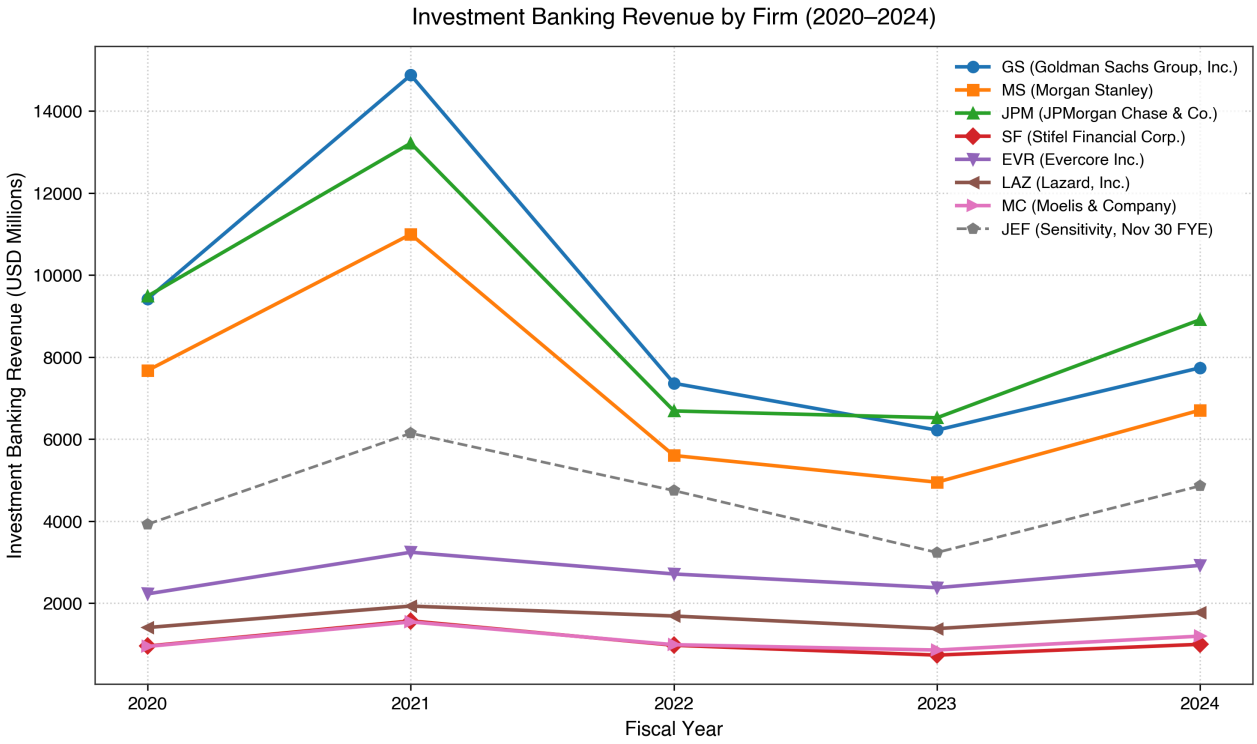


Figure 1: Investment-banking revenue by firm (USD millions, 2020 to 2024). Source: Form 10-K filings.

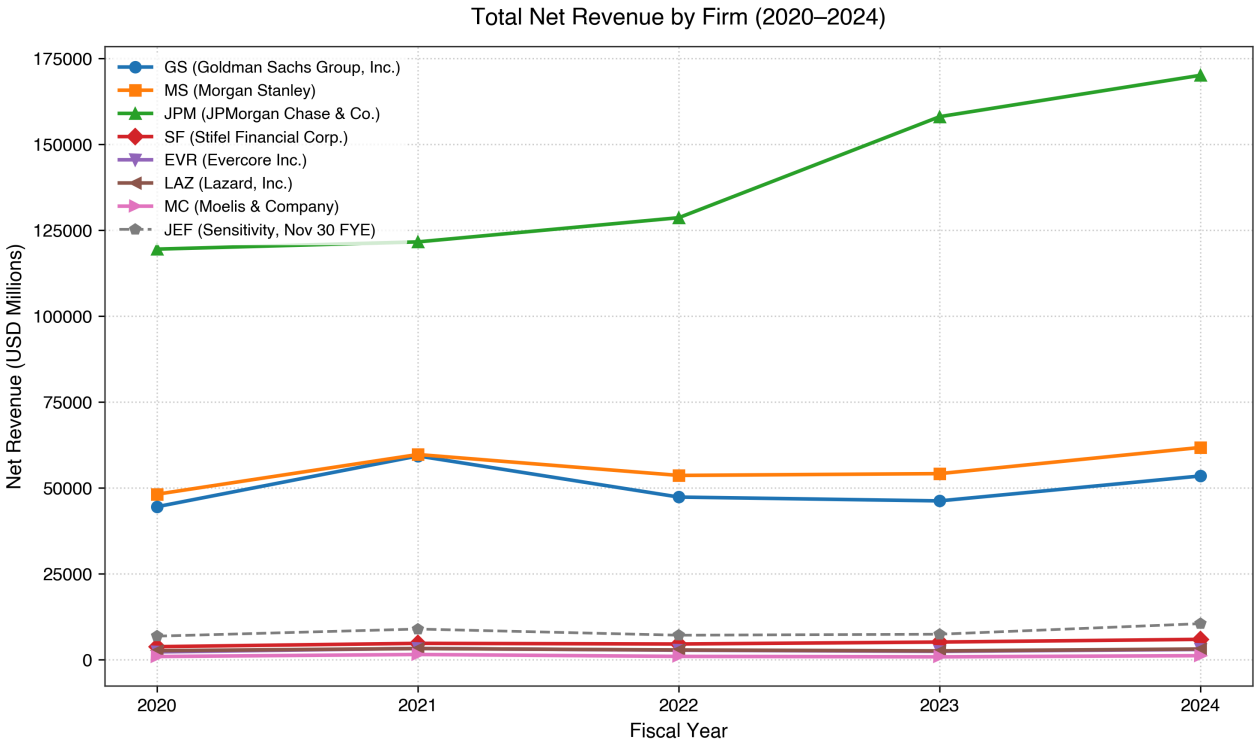


Figure 3: Net revenue by firm (USD millions, 2020 to 2024). Source: Form 10-K filings.

6.2 Dividends and buybacks

Over the five years the eight firms returned **$233.3 billion** to shareholders: $113.9 billion in cash dividends (48.8%) and $119.4 billion in share repurchases (51.2%).

Table 3: Shareholder distributions, 2020 to 2024 (USD millions)

Ticker	5-year dividends	5-year buybacks	5-year total	Dividend change 2020 to 2024	Yearly buyback range
GS	17,429.0	24,424.0	41,853.0	+92.5%	$1.9B to $8.0B
MS	24,212.0	31,226.0	55,438.0	+124.1%	$1.3B to $11.5B
JPM	67,356.0	56,741.0	124,097.0	+16.5%	$3.2B to $18.8B
SF	774.3	924.8	1,699.1	+208.0%	$58M to $444M
EVR	616.4	2,223.3	2,839.7	+27.4%	$147M to $721M
LAZ	926.5	1,354.6	2,281.1	-9.0%	$60M to $692M
MC	1,303.9	353.8	1,657.7	-34.9%	$11M to $148M
JEF	1,245.4	2,159.6	3,405.0	+88.3%	$44M to $860M

1. **Dividends** rose at all four banks: Goldman Sachs from $2,336M to $4,497M, Morgan Stanley from $2,739M to $6,138M, JPMorgan Chase from $12,690M to $14,783M and Stifel from $73.8M to $227.3M. Only JPMorgan Chase had a small dip (in 2023). The boutiques were mixed: Evercore's dividends rose, but Lazard's and Moelis's fell (Moelis paid $479.9M in 2021 and $174.7M in 2022).
2. **Buybacks** moved much more (Figure 2). In 2021, the strongest revenue year, they jumped at JPMorgan Chase ($6,517M to $18,408M), Morgan Stanley ($1,347M to $11,464M) and Goldman Sachs ($1,928M to $5,200M). After that the paths split. Morgan Stanley cut buybacks every year, to $3,250M in 2024. JPMorgan Chase fell to $3,162M in 2022 and then rose to $18,830M in 2024. Goldman Sachs bought back more in 2023 ($5,796M) and 2024 ($8,000M), even though its ROE fell to 7.5% in 2023. So buybacks were high in the boom year, but after that they did not simply follow revenue or profit.

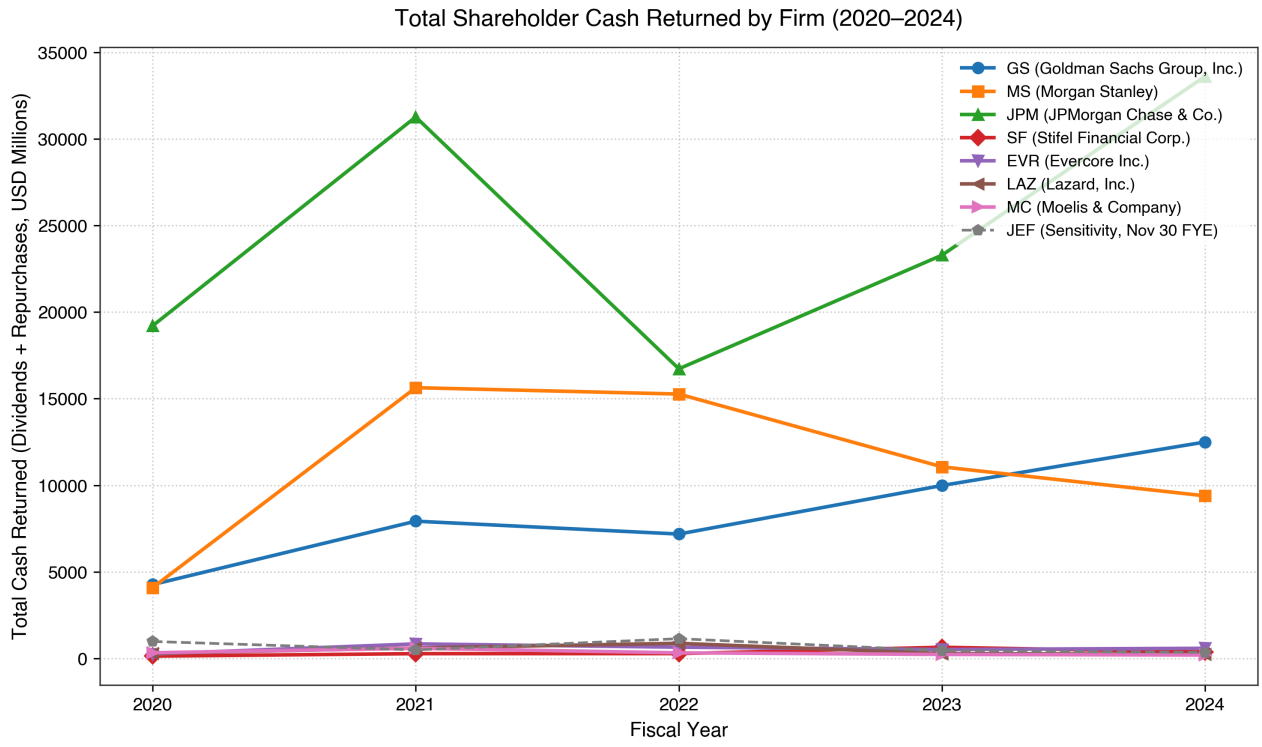


Figure 2: Total cash returned to shareholders by firm (USD millions, 2020 to 2024). Source: Form 10-K cash-flow statements.

6.3 Bank capital ratios and returns (Panel A)

Table 4: Panel A standardized CET1 ratio and ROE (%)

Ticker	Measure	2020	2021	2022	2023	2024	5-year mean
GS	CET1	14.7	14.2	15.0	14.4	15.0	14.66
GS	ROE	11.1	23.0	10.2	7.5	11.0	12.56
MS	CET1	17.4	16.0	15.3	15.2	15.2	15.82
MS	ROE	15.2	19.8	15.3	12.8	14.5	15.52
JPM	CET1	13.1	13.0	13.2	15.0	15.3	13.92
JPM	ROE	12.0	19.0	14.0	17.0	18.0	16.00
SF	CET1	10.9	11.2	11.3	11.4	11.4	11.24
SF	ROE	13.1	17.5	14.3	11.1	11.8	13.56

For Goldman Sachs, Morgan Stanley and JPMorgan Chase, CET1 ratios stayed between 13.0% and 17.4% (Figure 4). Morgan Stanley had the highest average (15.82%) but fell from 17.4% to 15.2%; JPMorgan Chase rose from 13.1% to 15.3%. Stifel, under Category IV rules, stayed between 10.9% and 11.4%.

Supplementary leverage ratios at the three G-SIBs were higher in 2020 (6.9% to 7.4%), when a temporary Federal Reserve rule left U.S. Treasuries and central bank deposits out of the calculation. After the rule expired on 31 March 2021, they were between 5.4% and 6.2%.

ROE peaked in 2021 at all four banks, from 17.5% at Stifel to 23.0% at Goldman Sachs (Figure 5). In 2023 it fell to 7.5% at Goldman Sachs, 12.8% at Morgan Stanley and 11.1% at Stifel, while JPMorgan Chase's rose to 17.0%. In 2024 it ranged from 11.0% to 18.0%.

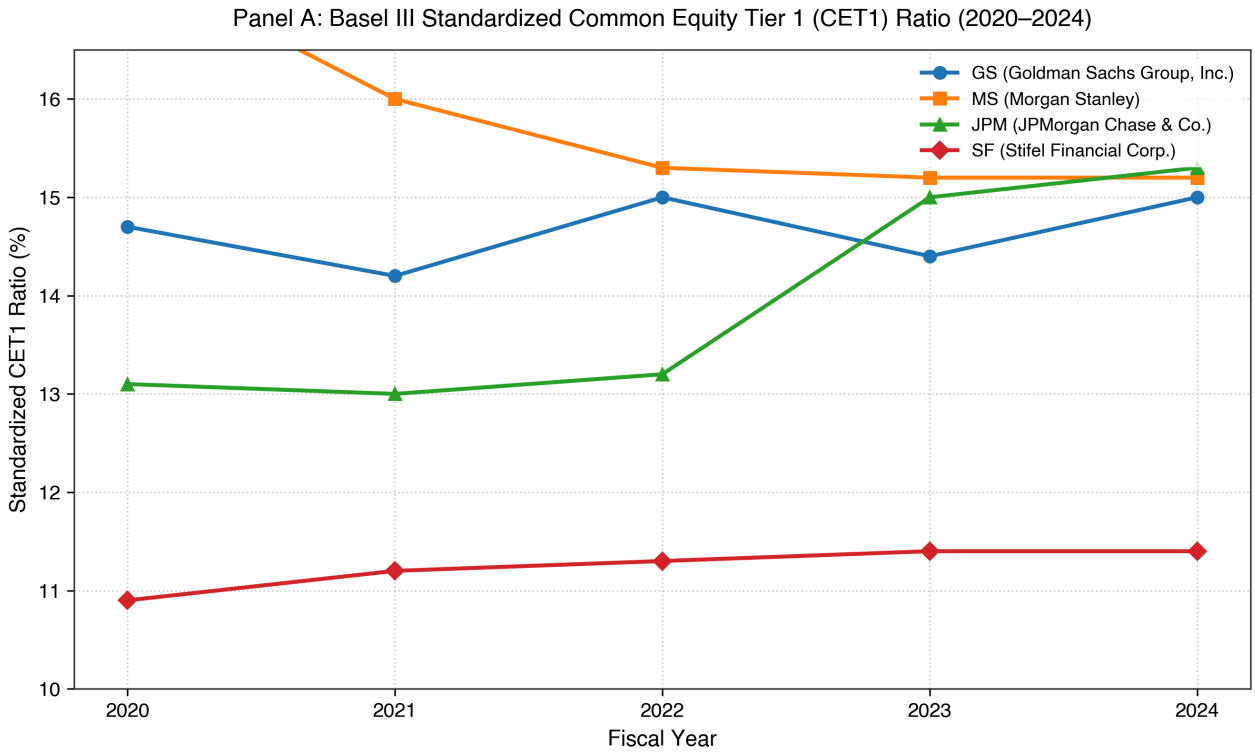


Figure 4: Standardized CET1 ratios for Panel A (2020 to 2024). Source: Form 10-K Item 7, capital resources.

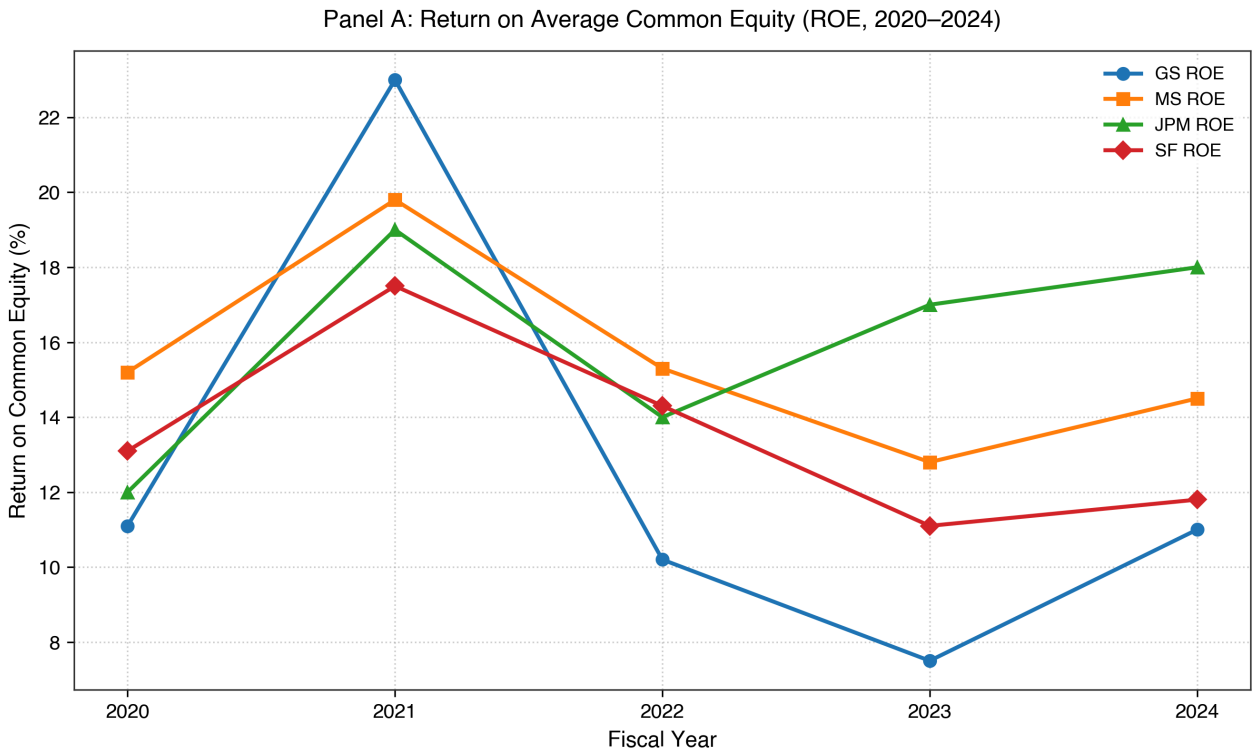


Figure 5: Return on common equity (%) for Panel A (2020 to 2024). Source: Form 10-K financial highlights.

6.4 Boutique compensation and margins (Panel B)

Table 5: Panel B compensation ratio and operating margin (%)

Ticker	Measure	2020	2021	2022	2023	2024	5-year mean
EVR	Compensation ratio	64.5	60.7	60.1	66.0	66.0	63.5
EVR	Operating margin	19.5	27.9	25.2	15.0	16.5	20.8
LAZ	Compensation ratio	61.3	61.1	61.1	71.6	65.0	64.0
LAZ	Operating margin	18.3	20.1	16.6	-3.4	10.9	12.5
MC	Compensation ratio	63.0	64.0	65.0	70.0	65.0	65.4
MC	Operating margin	24.6	30.2	21.7	10.6	19.7	21.4

Pay is the boutiques' biggest cost (Figure 6). In 2021 compensation ratios were 60.7% to 64.0%; in 2023 they rose to 66.0% at Evercore, 70.0% at Moelis and 71.6% at Lazard. The ratio rose mainly because pay did not fall as fast as revenue. From 2022 to 2023 compensation went from $1,671.5M to $1,612.2M at Evercore and from $640.4M to $598.3M at Moelis, and it rose from $1,745.3M to $1,855.9M at Lazard, while net revenue fell at all three.

Operating margins moved the other way (Figure 7): 20.1% to 30.2% in 2021, then 15.0% (Evercore), 10.6% (Moelis) and -3.4% (Lazard) in 2023, and 10.9% to 19.7% in 2024. Lazard had a GAAP operating loss of $87.7M in 2023, which I attribute to restructuring charges. So in the 2023 slowdown the boutiques kept paying their bankers at close to the same level, and the fall in revenue showed up mostly in lower margins.

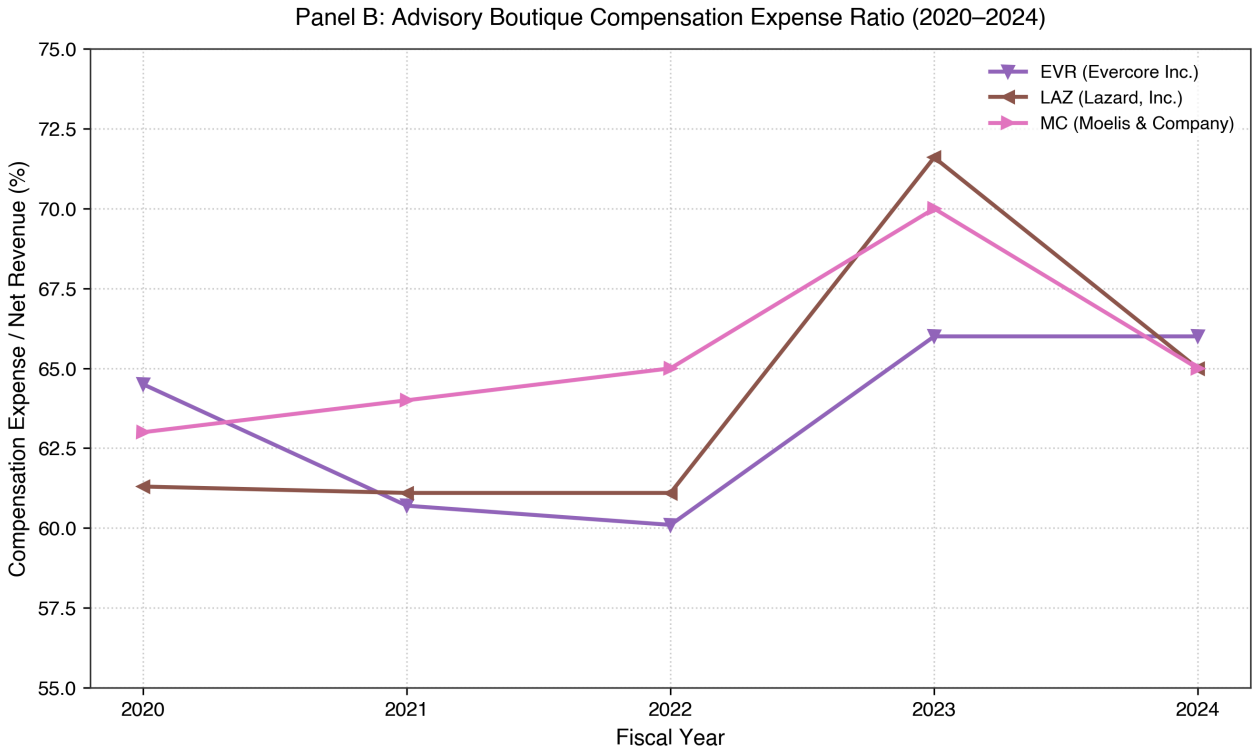


Figure 6: Compensation ratio (%) for Panel B (2020 to 2024). Source: Form 10-K statements of earnings.

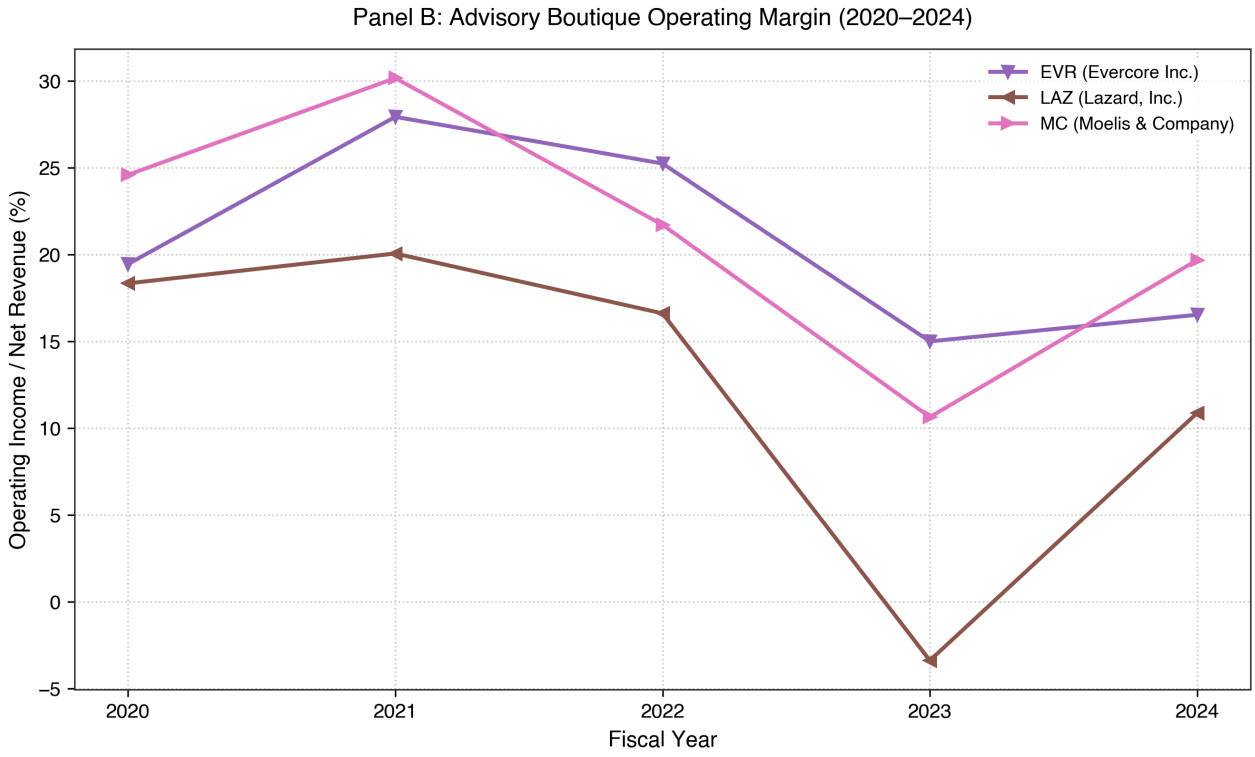


Figure 7: Operating margin (%) for Panel B (2020 to 2024). Source: Form 10-K statements of earnings.

6.5 Balance sheets

Evercore and Moelis had no funded long-term debt in any year; in 2024 Evercore held $687.4M in cash and Moelis $432.8M. Lazard carried senior notes averaging $1,682.2M against average cash of $1,222.5M, so its net liquid cushion was negative every year (between -$209.7M and -$760.6M). I think this reflects its asset-management business. The banks fund themselves with deposits, repurchase agreements and long-term debt as part of their lending and trading, so I did not compare their cash and debt with the boutiques'.

7. COMPARATIVE AND LAGGED ANALYSIS

7.1 Why the panels are analysed separately

For a bank, capital is the regulatory buffer that absorbs losses, and payouts are limited by the Federal Reserve's annual stress tests (CCAR). For a boutique, capital is mainly cash to pay staff through slow periods, and its board faces no bank capital rules. Banks also have large non-pay costs such as interest, credit provisions and technology, while boutique costs are mostly compensation. Pooling the two would mix these differences.

7.2 Lagged associations (t to t+1)

Each firm gives four pairs (2020 to 2021 through 2023 to 2024), so N = 16 for Panel A (4 firms x 4) and N = 12 for Panel B (3 firms x 4). Table 6 shows eight of the 30 pairs in the results file (`PHASE4_TABLES.md`, Table 8).

Table 6: Selected lagged associations (t to t+1)

Panel	Year t measure	Year t+1 outcome	N	Pearson r	p	Spearman rho	p
A	Share repurchases	ROE	16	-0.049	0.858	-0.035	0.897
A	Cash dividends paid	ROE	16	+0.266	0.319	+0.159	0.557
A	Cash dividends paid	Efficiency ratio	16	-0.483	0.058	-0.218	0.418
A	Standardized CET1	ROE	16	+0.174	0.520	+0.162	0.549
B	Share repurchases	Operating margin	12	-0.403	0.194	-0.385	0.217
B	Total cash returned	Operating margin	12	-0.371	0.234	-0.224	0.484
B	Compensation ratio	Operating margin	12	-0.036	0.912	-0.102	0.753
B	IB revenue	Operating income	12	+0.470	0.123	+0.503	0.095

7.3 Results

1. **Panel A, buybacks and next-year ROE**: close to zero (r = -0.049, rho = -0.035; Figure 8). In this sample, bigger buybacks were not followed by higher or lower ROE.
2. **Panel A, dividends and next-year efficiency ratio**: the strongest Pearson result in Panel A (r = -0.483, p = 0.058), but Spearman was much weaker (rho = -0.218, p = 0.418). When the two disagree this much, the Pearson value probably depends on a few large values, so I do not rely on it.
3. **Panel B, buybacks and next-year operating margin**: negative (r = -0.403, p = 0.194; rho = -0.385) but not significant. I think it reflects timing. Evercore's buybacks peaked in 2021 ($720.7M) and Lazard's and Moelis's in 2022 ($691.7M and $147.5M), and margins at all three fell in 2022 and again in 2023. This is a timing pattern, not evidence that buybacks lowered margins.
4. **Panel B, compensation ratio and next-year margin**: almost no link (r = -0.036; Figure 9).
5. **Panel B, advisory revenue and next-year operating income**: moderately positive (r = +0.470, p = 0.123; rho = +0.503, p = 0.095). A strong fee year tended to be followed by another fairly strong year, but the result is not significant at the 5% level.

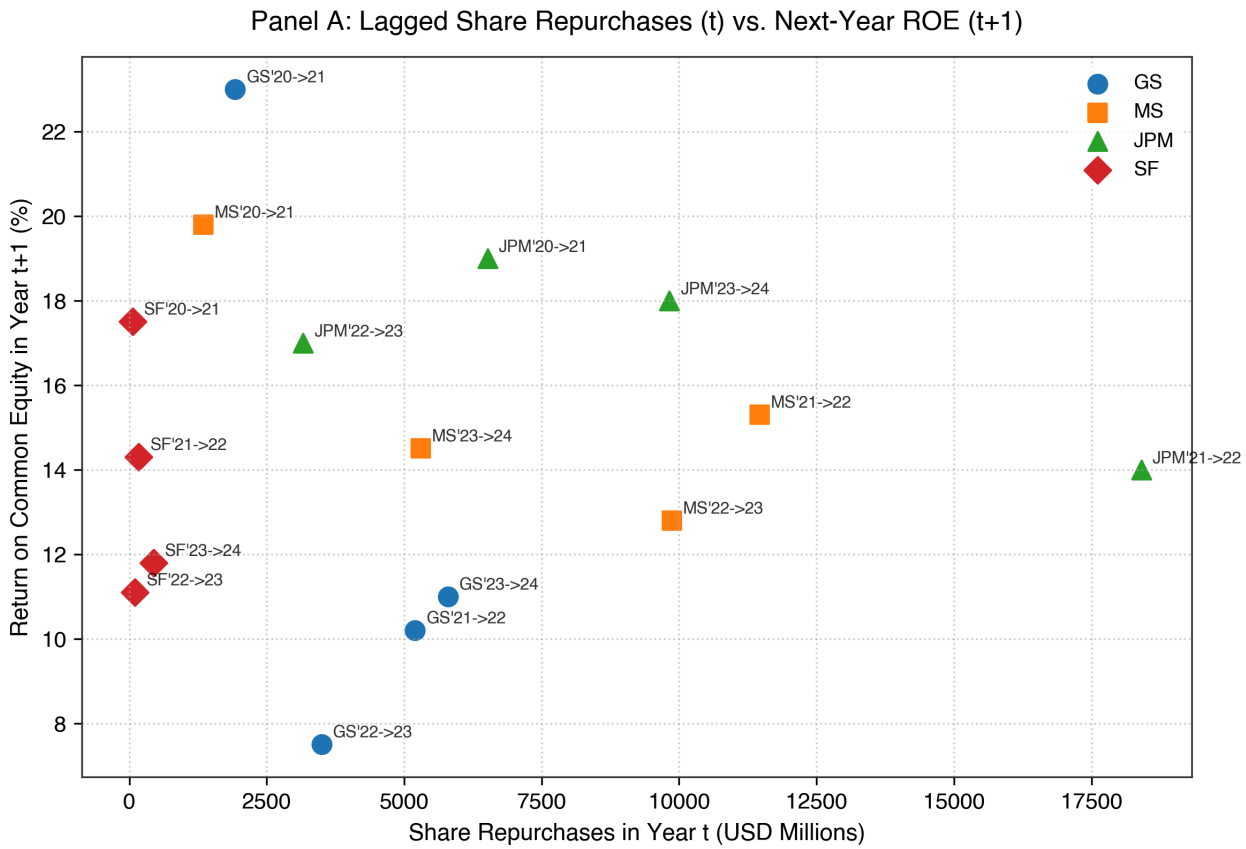


Figure 8: Share repurchases in year t against ROE in year t+1, Panel A (N = 16). Source: Form 10-K filings.

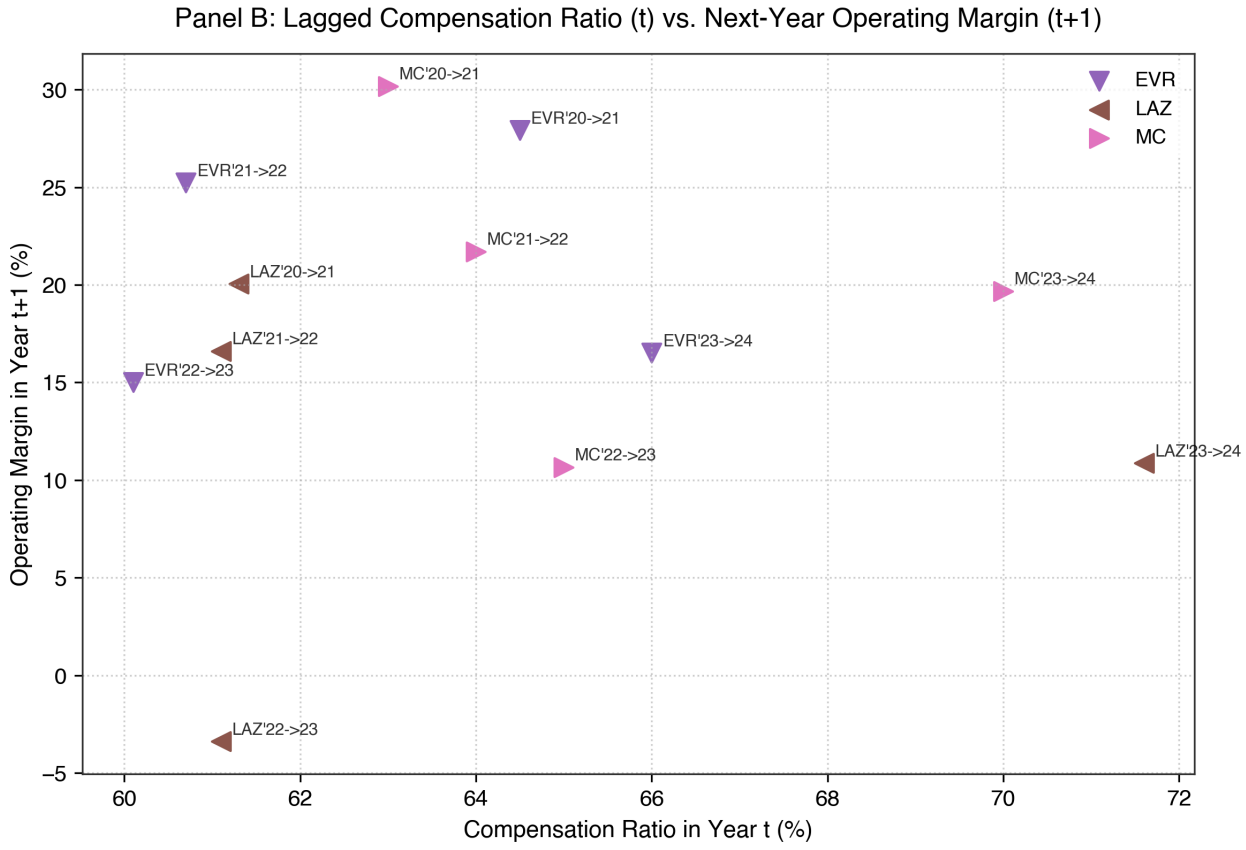


Figure 9: Compensation ratio in year t against operating margin in year t+1, Panel B (N = 12). Source: Form 10-K filings.

7.4 How to read these numbers

These correlations only describe how the numbers moved together over four transitions. Pairs from the same firm are not independent, so the p-values are a rough guide only.

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8. DISCUSSION

The two panels behaved differently. The data show almost no link between bank buybacks and next-year ROE (r = -0.049), and a weak negative link between boutique buybacks and next-year margin (r = -0.403). The most interesting point for me is not the sign of either result but that the same kind of measure behaved differently in the two groups. A buyback by a regulated bank and a buyback by a boutique are not the same decision. Market cycles may affect the timing of payouts and profits differently in each model, but I cannot test this.

Dividends and buybacks. Bank dividends rose between 2020 and 2024 (+16.5% to +208.0%), while buybacks jumped in 2021 and then went different ways at each firm. This is consistent with banks treating dividends as a regular payment they avoid cutting and buybacks as the flexible part. Firms may buy back shares to return extra capital, to offset shares issued to staff, or because they think their shares are cheap; the filings do not tell me which.

Capital and costs mean different things in each model. The three large banks kept CET1 ratios of 13.0% to 17.4%. Evercore and Moelis had no funded debt and held net cash of $336.8M to $687.4M, while Lazard had net debt every year. In 2023 boutique compensation ratios rose to 66.0% to 71.6% and margins fell to between -3.4% and 15.0%. Bank efficiency ratios moved less at JPMorgan Chase (53.0% to 62.0%) than at Goldman Sachs (53.8% to 74.7%). My interpretation is that at a bank, retained capital is mainly a regulatory buffer, and stress tests and capital rules limit how much it can pay out. At a boutique, cash is a reserve for slow years and pay is the main cost to manage. Banks also have several business lines with different cycles, while boutiques depend almost fully on advisory fees, which may explain why boutique profits fell more sharply in 2023. I did not measure these factors directly.

Decisions as an interpretive link. Considerations such as risk tolerance, keeping staff, liquidity needs and shareholder expectations probably sit between conditions and outputs (Section 3), but they are not in 10-K filings, so this is interpretation, not a finding. I do not rank the two business models.

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9. LIMITATIONS

1. **Small samples**: only four transitions per firm (N = 16 and N = 12), so all correlations are exploratory.
2. **Many tests**: the results file has 30 lagged pairs. Only one had a p-value below 0.05 (Panel B shareholder distribution ratio and next-year operating income, r = -0.614, p = 0.034), which is close to what chance alone would produce with 30 tests. I do not treat any single result as established.
3. **Short period**: five years cannot show long-term trends.
4. **Unusual years**: pandemic relief, near-zero rates, a record 2021 deal year and 525 basis points of rate rises. Normal years may look different.
5. **Observational design**: I track reported numbers, so I cannot claim causes.
6. **No data on decisions or motives**: filings report results, not board discussions, and cannot tell me why a firm bought back shares.
7. **No deal-level data**: I cannot test whether deals that run across two years explain the lagged results.
8. **Common market movements**: revenue moved the same way at most firms, so I cannot separate market-wide effects from each firm’s choices.
9. **Differences inside panels**: JPMorgan Chase has a large deposit business, Stifel a wealth-management network and Lazard an asset-management arm.
10. **One-year lag**: some capital decisions may take several years to affect results.

10. CONCLUSION

Across 40 Form 10-K filings, the eight firms returned $233.3 billion to shareholders, bank dividends rose while buybacks changed a lot, the three largest banks kept CET1 ratios of 13.0% to 17.4%, and the boutiques' 2023 slowdown showed up in lower margins because pay held up while revenue fell. Bank buybacks had almost no link with next-year ROE (r = -0.049), and boutique buybacks only a weak, non-significant negative link with next-year margin (r = -0.403).

My main takeaway is that a measure such as buybacks or capital should be read in the context of the firm's business model, balance sheet and regulation. The study cannot show causes, observe motives, be generalised with confidence or say what the best capital-allocation strategy is.

11. FUTURE RESEARCH

Useful next steps would be a longer period (about 2005 to 2025, to include the 2008 crisis), quarterly 10-Q data, European banks such as Barclays, Deutsche Bank and UBS under the European Central Bank and Bank of England, deal-level timing data, market indicators such as interest-rate spreads and the VIX, and investment-banking revenue split into M&A, equity underwriting, debt underwriting and restructuring.

12. REFERENCES

Primary regulatory and statutory sources

1. **U.S. Securities and Exchange Commission (SEC)**:
   - Form 10-K Annual Reports (2020, 2021, 2022, 2023, 2024) for:
     - The Goldman Sachs Group, Inc. (CIK: 0000886982)
     - Morgan Stanley (CIK: 0000895421)
     - JPMorgan Chase & Co. (CIK: 0000019617)
     - Stifel Financial Corp. (CIK: 0000720672)
     - Evercore Inc. (CIK: 0001360901)
     - Lazard, Inc. / Lazard Ltd (CIK: 0001311370)
     - Moelis & Company (CIK: 0001591890)
     - Jefferies Financial Group Inc. (CIK: 0000096223)
   - SEC Rule 15c3-1: *Net Capital Requirements for Brokers or Dealers*.
2. **Board of Governors of the Federal Reserve System**:
   - *Regulatory Capital Rules: Implementation of Risk-Based Capital Standards (Basel III)*, 12 C.F.R. Part 217.
   - *Framework for Prudent Supervision of Large Bank Holding Companies*, Category I through Category IV Banking Organizations.
   - *Temporary Exclusion of U.S. Treasury Securities and Deposits at Federal Reserve Banks from the Supplementary Leverage Ratio*, Interim Final Rule (April 2020; expired March 31, 2021).
   - *Comprehensive Capital Analysis and Review (CCAR) and Stress Testing Regulations*, 12 C.F.R. Part 252.
3. **Basel Committee on Banking Supervision (BCBS)**:
   - *Basel III: A global regulatory framework for more resilient banks and banking systems*, Bank for International Settlements (BIS).

13. APPENDIX

Appendix A: Variable definitions

- **Net revenue**: net revenues after interest expense, from the income statement.
- **Investment-banking revenue**: financial advisory, equity underwriting and debt underwriting revenue.
- **Cash dividends paid** and **share repurchases**: cash amounts from financing cash flows (repurchases include shares withheld for employee tax; see Section 5).
- **Standardized CET1 ratio**: Common Equity Tier 1 capital / standardized risk-weighted assets under Basel III.
- **Compensation ratio**: compensation and benefits expense / net revenue.
- Other derived measures are defined in Section 5.

Appendix B: Provenance

Each raw value is linked to its SEC EDGAR accession number, statement line and filing date in `PHASE3_PROVENANCE_MAP.csv` and `PHASE3_SOURCE_REGISTER.csv`.

Appendix C: Statistical notes

- **Pearson r** = sum((x - mean of x)(y - mean of y)) / sqrt(sum((x - mean of x)^2) x sum((y - mean of y)^2))
- **Spearman rho** = 1 - 6 x sum(d^2) / (n(n^2 - 1)), where d is the difference between the ranks of x and y in each pair.
- I compared the direction and size of Pearson r and Spearman rho for each pair to see whether a result depended on a few extreme values.